

RecoMart Data Quality Report

Run ID: 20260429T075431Z
Generated at: 2026-04-29T07:54:31.997715+00:00
Project Root: C:\Users\barath\recomart-pipeline

1. Initial Validation

Initial overall status: **FAIL**

Initial Dataset Summary

phase	dataset	rows	columns	errors	warnings	status
initial	retailrocket_events	2756101	5	0	2734104	PASS
initial	retailrocket_category_tree	1669	2	0	25	PASS
initial	retailrocket_item_properties	20275902	4	0	0	PASS
initial	dummyjson_products_raw	194	27	0	92	PASS
initial	dummyjson_categories_raw	24	3	1	0	FAIL
initial	products_parquet	194	32	0	0	PASS
initial	categories_parquet	24	8	0	0	PASS

Initial Validation Issues

	check_type	severity	issue_count	description	file_path
	missing_values	warning	2733644	Nulls in transactionid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\lo
	duplicates	warning	460	Duplicate rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\lo
ee	missing_values	warning	25	Nulls in parentid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-26\lo
aw	missing_values	warning	92	Nulls in brand	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\lo
_raw	schema	error	1	Missing columns: ['category']	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\lo

2. Data Fixing Actions

The following remediation steps were applied automatically to failed or warning-prone datasets.

Fix Log

	rows_before	rows_after	details	file_path
event_values	2756101	2756101	Kept only allowed event types	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
estamps	2756101	2756101	Removed rows with invalid timestamp	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
	2756101	2755641	Removed duplicate event rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
quired_rows	2755641	2755641	Dropped rows missing required fields	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
categoryid	1669	1669	Dropped rows with missing categoryid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
	1669	1669	Removed duplicate category rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
quired_rows	20275902	20275902	Dropped rows with invalid required values	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
	20275902	20275902	Removed duplicate property rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
quired_rows	194	194	Dropped rows missing id/title/category/price	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
ds	194	194	Kept first product per id	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
es	194	194	Clipped 0 negative prices to 0	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
	194	194	Clipped 0 rating values into 1-5	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
ck	194	194	Clipped 0 stock values to 0	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
ategory	24	0	Dropped null categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
gory	0	0	Dropped blank categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
	0	0	Removed duplicate categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
onze_metadata	194	194	Ensured bronze metadata columns exist	C:\Users\barath\recomart-pipeline\data\bronze\dummyjson\products.pa
onze_metadata	24	24	Ensured bronze metadata columns exist	C:\Users\barath\recomart-pipeline\data\bronze\dummyjson\categories.p

3. Revalidation

Revalidation overall status: **PASS**

Revalidated Dataset Summary

phase	dataset	rows	columns	errors	warnings	status
revalidated	retailrocket_events	2755641	5	0	2733184	PASS
revalidated	retailrocket_category_tree	1669	2	0	25	PASS
revalidated	retailrocket_item_properties	20275902	4	0	0	PASS
revalidated	dummyjson_products_raw	194	27	0	92	PASS
revalidated	dummyjson_categories_raw	0	1	0	0	PASS
revalidated	products_parquet	194	32	0	0	PASS
revalidated	categories_parquet	24	8	0	0	PASS

Revalidation Issues

dataset	check_type	severity	issue_count	description	file_path
retailrocket_events	missing_values	warning	2733184	Nulls in transactionid	C:\Users\barath\recomart-pipeline\data\remediated\raw\events.
retailrocket_category_tree	missing_values	warning	25	Nulls in parentid	C:\Users\barath\recomart-pipeline\data\remediated\raw\category.
dummyjson_products_raw	missing_values	warning	92	Nulls in brand	C:\Users\barath\recomart-pipeline\data\remediated\raw\product.

4. Before vs After

Validation Comparison

metric	before	after
overall_status	FAIL	PASS
error_count	1	0
warning_count	2734221	2733301

5. Conclusion

The initial validation run returned **FAIL**. After applying automated remediation steps and rerunning validation on the remediated files, the final status returned **PASS**. The detailed issue logs, fix log, and dataset summaries have been saved alongside this PDF.